

Open Sinhala & Tamil corpora for speaker verification

Every openly-licensed corpus that could widen the si/ta benchmark — what each one actually gives you, and which three are worth the download.

2026-08-10 survey EXECUTED — see section 05 research_logs/2026-08-10-open-si-ta-sv-corpora.md

Read section 05 first if you are acting on this. Sections 00-04 are the survey as written *before* anything was downloaded. Five corpora were then built, and doing so corrected six claims below — each marked CORRECTED inline. Nothing was deleted, so the original reasoning stays auditable.

00 The short version

Three findings shape every recommendation below.

FINDING 1

Tamil is the binding constraint — and it is fixable

49 speakers cannot resolve a 0.3 % EER delta. Two corpora take the Tamil pool past 750 speakers, and one of them ships ASV trial lists already built.

FINDING 2

Sinhala has almost nothing new

SLR52 is the open Sinhala corpus. The only genuinely new audio is SiTa (10 h, in-the-wild) and SLCeleb. Sinhala is absent from Common Voice entirely.

FINDING 3

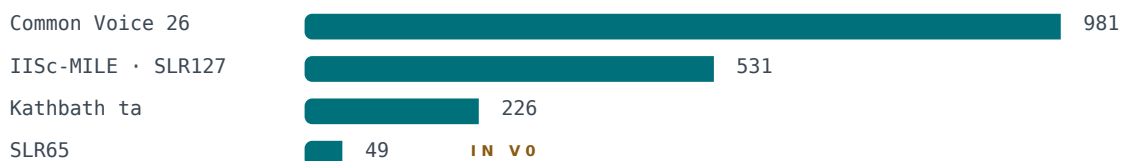
Every large Tamil corpus is Indian Tamil

SLR65, SLR127, Kathbath, IndicVoices, Vaani — all recorded in India. Excellent training data; invalid for a Sri Lankan Tamil performance claim.

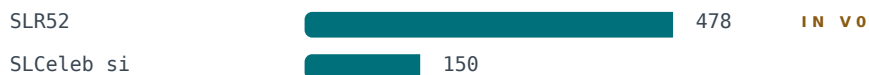
Speakers available per open corpus

Distinct speakers with usable identity labels. Corpora whose speaker counts are undocumented are omitted rather than estimated.

TAMIL



SINHALA



0 — 981 speakers · bars share one scale across both languages

Sinhala has one open option and it is already in the benchmark. Tamil has three unused ones, each larger than what v0 runs on.

01 Use these

Open licence, speaker identity recoverable, enough speakers to move the numbers.

Kathbath / IndicSUPERB

AI4Bharat

ta (+11 Indic) · 226 Tamil speakers · 1,684 h all langs

CC0

The only corpus here with a **ready-made ASV protocol** — valid_data.txt, test_known_data.txt, test_data.txt per language. Speaker and gender sit in the filenames. Ships a **noisy test variant**, which is a free robustness axis for the ablation. Audio is m4a. **[CORRECTED 05: the 226 counts the 85 GB train split too; the three eval splits hold 20 speakers each (60 built), so the trial count is large while the speaker count is not.]**

IISc-MILE Tamil ASR Corpus

OpenSLR SLR127

ta · 531 speakers · ~150 h

CC BY 2.0

Biggest open Tamil speaker pool. Clean read speech, 16 kHz / 16-bit mono, train/test split already present. **Caveat:** the OpenSLR page does not document the speaker-ID convention — inspect filenames before budgeting the work.

Large Sinhala ASR training set

OpenSLR SLR52

si · 478 speakers (≥ 10 utts) · ~185k utts

CC BY-SA 4.0

Already ingested and still the Sinhala backbone. Crowdsourced by Google in Sri Lanka; speaker IDs come from utt_spk_text.tsv. No gender, no session metadata.

SLCeleb

ours · IEEE DataPort

si + ta · 280 speakers · 34k utts

CC BY 4.0

Still the only source of multi-session, multi-genre, bilingual, Sri-Lankan-dialect speakers. Remains the v1 blocker — the audio is not on the local mounts.

02 Worth adding, with caveats

Real value, but each carries a condition that changes how you can use it.

Common Voice 26.0

Mozilla · released 2026-06-12

ta only · 981 distinct voices · 425 h / 235 h validated

CC0

client_id is a pseudo-speaker ID and contributors record *across multiple days*. **[CORRECTED 05: calling this a genuine cross-session source was overstated — CV ships no session field and no timestamp, so cross-sitting pairs are likely but unenforceable. Its real distinction is device and channel diversity.]** Gender is missing on ~66 % of clips, which constrains same-gender impostor sampling. No Sinhala.

IndicVoices / IndicVoices-R

AI4Bharat · NeurIPS 2024

ta (+21) · 10,496 speakers all langs · 1,704 h

CC BY 4.0

Spontaneous and read speech with rich per-speaker metadata — pitch, SNR, C50, demographics. Tamil subset also mirrored on Kaggle. Per-language speaker counts need checking before you plan around them.

Vaani

ARTPARK-IISc × Google DeepMind

ta (+104) · 158,441 speakers · ~31,255 h raw

CC BY 4.0

Image-prompted spontaneous speech, downloadable **per district** — so you can pull Tamil Nadu only. The authors explicitly pitch it for speaker ID/verification. Far too large to be an eval set; treat it as a pretraining and AS-Norm cohort pool.

NISP

IISc LEAP

ta (+4 Indic +en) · 345 speakers · ~4–5 min per speaker

open · GitHub

Small, but **every speaker records in both their mother tongue and English** — a real cross-lingual trial list without waiting on SLCeleb. Directly relevant to the P4 cross-lingual study.

SiTa

Univ. of Moratuwa · CHiPSAL 2025

si + ta · 1–10 spk/video · 602 min si, 121 min ta

CC BY-NC 4.0

The only new **Sri Lankan** in-the-wild audio — YouTube panel shows, debates, quizzes, code-mixed, real overlap. But the labels are *diarization turns*: speaker identity holds within a recording, not across them. Gives you conversational eval segments, not cross-recording trials, unless you link identities by hand. **[CORRECTED 05: licence is CC BY-NC 4.0 — NON-COMMERCIAL, so unusable in the VoiceID product path; access is also gated behind a request form.]**

SPRING-INX

SPRING Lab, IIT Madras

ta (+9) · speaker count not stated · ~2,000 h

CC BY 4.0

Large and legally clean, manually transcribed. Whether speakers are labelled well enough for SV needs verifying first.

Sinhala TTS

OpenSLR SLR30 · Google

si · multi-speaker, count unconfirmed · ~699 MB

CC BY-SA 4.0

UserID is embedded in the FileID, so speaker labels do exist. Almost certainly only a handful of speakers. Cheap to check, low expected yield.

Tamil multi-speaker set

OpenSLR SLR65 · Google

ta · 49 speakers (≥10 utts) · 4,286 utts

CC BY-SA 4.0

Already ingested. Keep it in the mix purely for continuity with the v0 numbers.

03 Checked and rejected

Ruled out on purpose, so nobody re-searches these.

Common Voice Sinhala

Does not exist. Not a size problem — the locale is absent from the corpus. Verified against the CV 26.0 locale list: 294 locales, no si.

VoxLingual107 si / ta

67 h Sinhala and 51 h Tamil under CC BY 4.0, but the segments are cut by *automatic* diarization — no reliable speaker identity. Fine for SSL or domain pretraining of the frontend; invalid for trials.

FLEURS si_lk / ta_in

Few speakers, no official speaker labels.

DISPLACE 2023 / 2024

Conversational Tamil with speaker labels from IISc — the closest thing to what we actually want — but access requires a signed Terms & Conditions submission, so it is not openly available. Worth requesting separately.

Microsoft Speech Corpus (Indian)

Tamil, Telugu, Gujarati. Research-only, non-commercial licence with mandatory attribution. Usable for a paper, not for the VoiceID product path.

IARPA Babel Tamil · LDC2017S13

Conversational telephone Tamil, the classic SV-style data — but LDC-licensed and paid. Noted only for completeness.

Sinhala TTS sets

SafnasKaldeen (HF), pnfo/sinhala-tts-dataset, SinhalaVITS — one to four speakers each. Useless for speaker verification.

04 What this changes for the ablation

The feature-combination study is currently underpowered on Tamil.

With 49 speakers the impostor pool is small enough that the EER confidence interval swamps the effects you are chasing — the P3 seed spread was already ± 0.10 EER on Tamil. Widening the corpus is not a nice-to-have before the ablation; it is a precondition for the ablation meaning anything.

- 1 **[DONE] Kathbath Tamil first.** CC0, ASV lists already exist, and the noisy variant doubles as a robustness condition. Lowest effort per unit of statistical power.
- 2 **[DONE] SLR127 second.** Biggest speaker count, and it is the same read-speech domain as v0, so it drops into the existing protocol once the speaker-ID convention is confirmed. (*Confirmed — and it turned out to carry real sessions; see section 05.*)
- 3 **[OPEN] Common Voice Tamil third. [CORRECTED 05: not for cross-session trials — Common Voice cannot provide those, and SLR127 already did. Fetch it for device and channel diversity instead. Needs a manual browser download.]**

- 4 **[OPEN] SiTa and SLCeleb as the held-out Sri Lankan set.** Never train on these. Both scaffolded; both still gated on external access.

Ingestion cost is small. `tools/ingest_openslr.py` is already structured as one `collect_<corpus>()` per source, so each new corpus is roughly a 25-line function plus a CLI flag. `tools/sl_dataprep.py` and the trial-list logic stay as they are.

The dialect split is a publishable axis, not just a caveat. Train on Indian Tamil (SLR127 + Kathbath), evaluate on Sri Lankan Tamil (SLCeleb / SiTa), report the gap. Nobody has published that number.

05 What was actually built

The section 04 plan was executed the same day. Five corpora are now prepared under `data/`, one self-contained folder each.

FOLDER	LANG	SPEAKERS	UTTS	HOURS	TRIALS
slr52_sinhala	si	478	185,293	224.5	12,000
slr127_tamil	ta	531	89,401	150.1	12,000 cross-session
kathbath_tamil	ta	60	7,171	13.2	150,000 official
nisp_tamil	ta + en	65	4,905	13.3	8,000 + 4,000 x-lingual
slr65_tamil	ta	50	4,291	7.1	12,000

291,061 files converted to 16 kHz mono PCM_16, **zero conversion errors**. All five load through the trainer's own `loadWAV`. Scaffolded but awaiting gated data: `commonvoice_tamil`, `sita_sinhala_tamil`, `slceleb_sinhala_tamil`. **The headline claim held: the Tamil speaker pool went 50 → 706.**

Corrections the build forced

SLR127 filename convention

Resolved by measurement: field 2 has exactly 531 distinct values, matching the documented speaker count. Field 3 is a *prompt* id reused by up to 312 speakers — it would have collided constantly as an utterance key.

SLR127 has genuine sessions — the survey missed this

Its three filename prefixes are collection batches, and 107 of 531 speakers span two of them. Verified: 0 same-session targets, all 107 contributing, 0 impostor pairs sharing a speaker. The only read-speech corpus here that can do this.

Common Voice cross-session claim was overstated

No session field, no timestamp. Cross-sitting pairs are likely but unenforceable.

SiTa is CC BY-NC 4.0

Non-commercial rules it out of the VoiceID product path — the only such restriction in the collection, and easy to forget later.

Kathbath's 226 speakers include the train split

The eval splits are 20 speakers each. Its 150,000 trials are powered by 20 speakers, not by 150,000 pairs.

NISP has two download traps

Tamil and English recordings sit in different GitHub folders — fetching only the first silently discards the entire reason to use NISP — and the concatenated archive is double-gzipped.

Do not regenerate Kathbath's trials. They are AI4Bharat's, translated to our paths (all 150,000, zero dropped). Resampling them destroys comparability with published IndicSUPERB numbers.

SiTa deliberately gets no trial list. Its RTTM speaker labels are recording-local, so any list built from it carries the same-recording confound — it would look like a benchmark and report flattering numbers.

SLCeleb is still the binding blocker. Only a one-speaker sample is on the mounts. It is the only corpus whose absolute EER would mean what a reader assumes, and the Indian-trained / Sri-Lankan-evaluated dialect gap is still the unpublished number.

06 Sources

OpenSLR index (SLR30, 52, 65, 127)	openslr.org/resources.php
IISc-MILE Tamil ASR	openslr.org/127
IndicSUPERB / Kathbath	github.com/AI4Bharat/IndicSUPERB · arXiv:2208.11761
Kathbath data	huggingface.co/datasets/ai4bharat/Kathbath
IndicVoices-R	github.com/AI4Bharat/IndicVoices-R · arXiv:2409.05356
Vaani	vaani.iisc.ac.in · huggingface.co/datasets/ARTPARK-IISc/Vaani
Common Voice stats	github.com/common-voice/cv-dataset — cv-corpus-26.0-2026-06-12.json
NISP	github.com/iiscleap/NISP-Dataset · arXiv:2007.06021
SiTa	aclanthology.org/2025.chipsal-1.8 · github.com/SiT-Ta-SpeakerDiarization/SiT-Ta
SPRING-INX	arXiv:2310.14654
VoxLingua107	cs.taltech.ee/staff/tanel.alumae/data/voxlingua107
DISPLACE	displace2024.github.io
Microsoft Speech Corpus	microsoft.com/en-us/download/details.aspx?id=105292